Scup (*Stenotomus chrysops*)

Summary Scores

- **Sensitivity:** Moderate
- **Exposure:** Moderate
- **Vulnerability:** Moderate

Sensitivity

Sensitivity Attributes	Score	LH DQ	Tally
Adult Mobility	1.52	2.9	
Complexity in Reproductive Strategy	2.92	2.9	
Dispersal of Early Life Stages	2.88	2.5	
Egg and Larval Survival and Settlement Requirements	2.24	2.5	
Historical Variability in Temperature	1.64	3	
Juvenile Habitat Requirements	1.32	3	
Other Stressors	2.76	2.8	
Population Growth Rate	2.4	3	
Prey Specificity	1.36	3	
Relationship to Calcified Organisms (due to impacts from Ocean Acidification)	1.28	2.9	
Spawning Cycle	3	2.9	
Stock Size/Status	1.12	3	
Total Sensitivity | Certainty	**Moderate (2) | 0.59**		

Exposure

Exposure Factors	Mean	SD	Tally
Bottom Temperature	2.4	1.19	
Bottom Oxygen	2.47	1.13	
Bottom Salinity	3.1	0.61	
Bottom Aragonite Solubility	1.32	0.81	
Net primary production	1.17	0.30	
Downward Flux of Particulate Organic Carbon	1.3	0.40	
Total Exposure	**Moderate (2.08)**	**1.02**	

Vulnerability

Vulnerability	Mean	SD	Tally
Total Vulnerability	**Moderate (4.15)**	**2.03**	

Low Moderate High Very High

Figure 1: Summary Sensitivity, Exposure, and Vulnerability Scores and Score Distributions for Scup. Exposure and Vulnerability scores are averaged spatially across the species distribution and within stocks if applicable (see Figure 5 and Figure 4). Life History Data Quality (LH DQ) and Standard Deviations (SD) are reported for senstivity and exposure/vulnerability scores, respectively.

Changes from Previous Assessment(s)

Table 1: Changes to Vulnerability, Exposure, and Sensitivity scores relative to Previous Assessments. If a citation for the previous assessment is not provided, it comes from Hare et al. (2016).

Metric	Previous Assessment(s)	Current Assessment	Change
Sensitivity	Low	Moderate	Increase
Exposure	Very High	Moderate	Decrease
Vulnerability	Moderate	Moderate	None
Directional Effect	Positive	Neutral	Decrease

Stocks

Scup is managed as a single stock or stock information was not considered for this iteration of the CVA.
Score Descriptions:

Sensitivity:

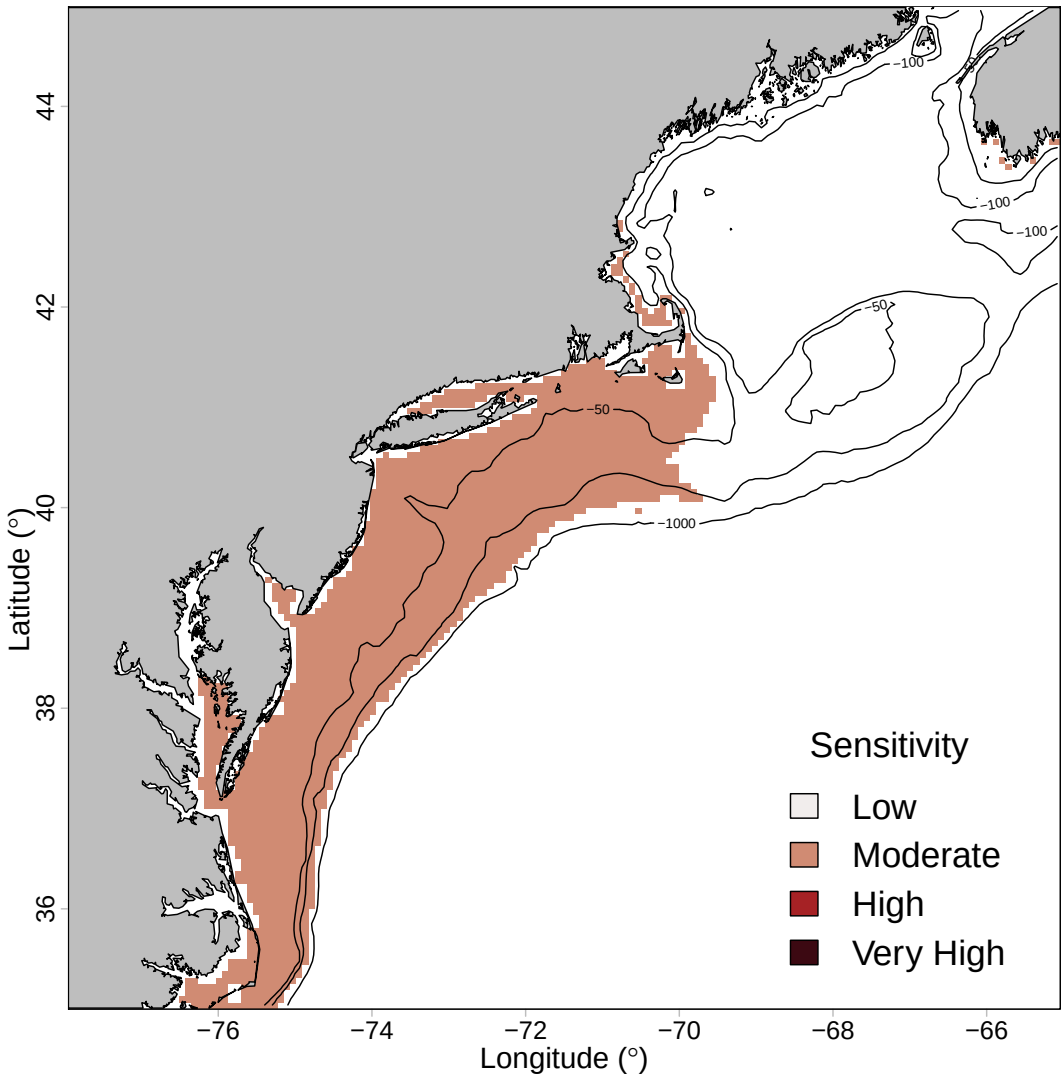


Figure 2: Scup Sensitivity on the Northeast Shelf

Scup received a **Moderate** sensitivity score with a certainty of 59%. A total of four attributes had scores greater than 2.5: Spawning Cycle (3), Complexity in Reproductive Strategy (2.9), Dispersal of Early Life Stages (2.9), and Other Stressors (2.8) (Figure 1).

Scup only spawn once per year between May and August (Steimle et al. 1999). Spawning occurs in inshore waters once they are warmer than 10C and rely on highly vulnerable habitats such as submerged aquatic vegetation and estuarine habitats (Steimle et al. 1999) (Guyant et al. 2022). Spawning may also rely on soniferous behaviors (Looby et al. 2022). Eggs and larvae do not disperse far from spawning locations (Steimle et al. 1999). Adults utilize benthic habitats on the continental shelf and in estuaries. As a result, they are exposed to a variety of stressors due to urbanization impacts such as hypoxia, changes to freshwater inputs, and dredging (Guyant et al. 2022).

In the 2016 Climate Vulnerability Assessment, Scup received a Low sensitivity score. Similar to our analysis, this score was driven by Spawning Cycle and Early Life History Requirements, as well as

Population Growth Rate (Hare et al. 2016). In our analysis, Population Growth Rate received a score of 2.4 (Figure 1).

Exposure:

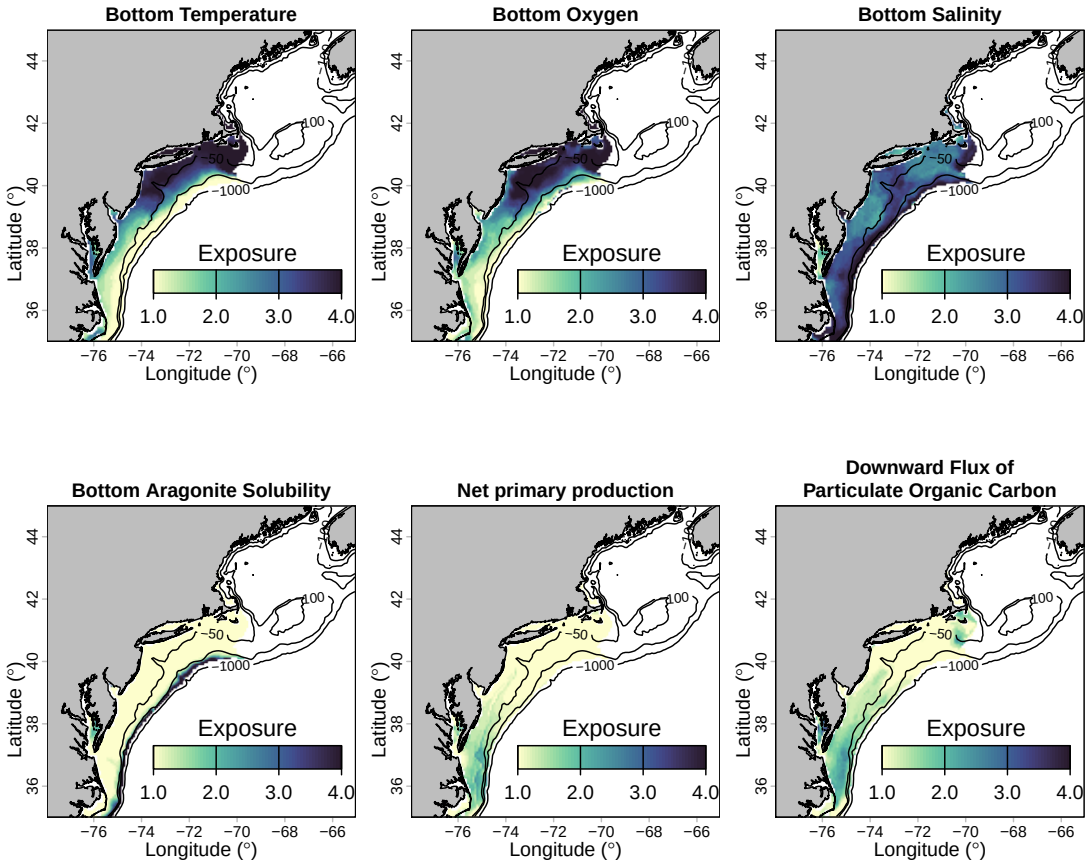


Figure 3: Maps of each Exposure Factor for Scup. Exposure was calculated between 2014-2023 and 2025-2035

Mean Exposure for Scup was **Moderate** (2.08 +/- 1.02) across its range (Figure 1). Exposure was Very High south of Cape Cod and High along the continental shelf throughout Southern New England. South of New Jersey, Exposure was Low, except along the continental shelf break (Figure 4). These patterns were driven by bottom temperature, oxygen, salinity, and aragonite solubility (Figure 3).

Scup received a Very High Exposure score in the previous Assessment. This score was driven by sea surface temperature, air temperature, and ocean acidification (Hare et al. 2016).

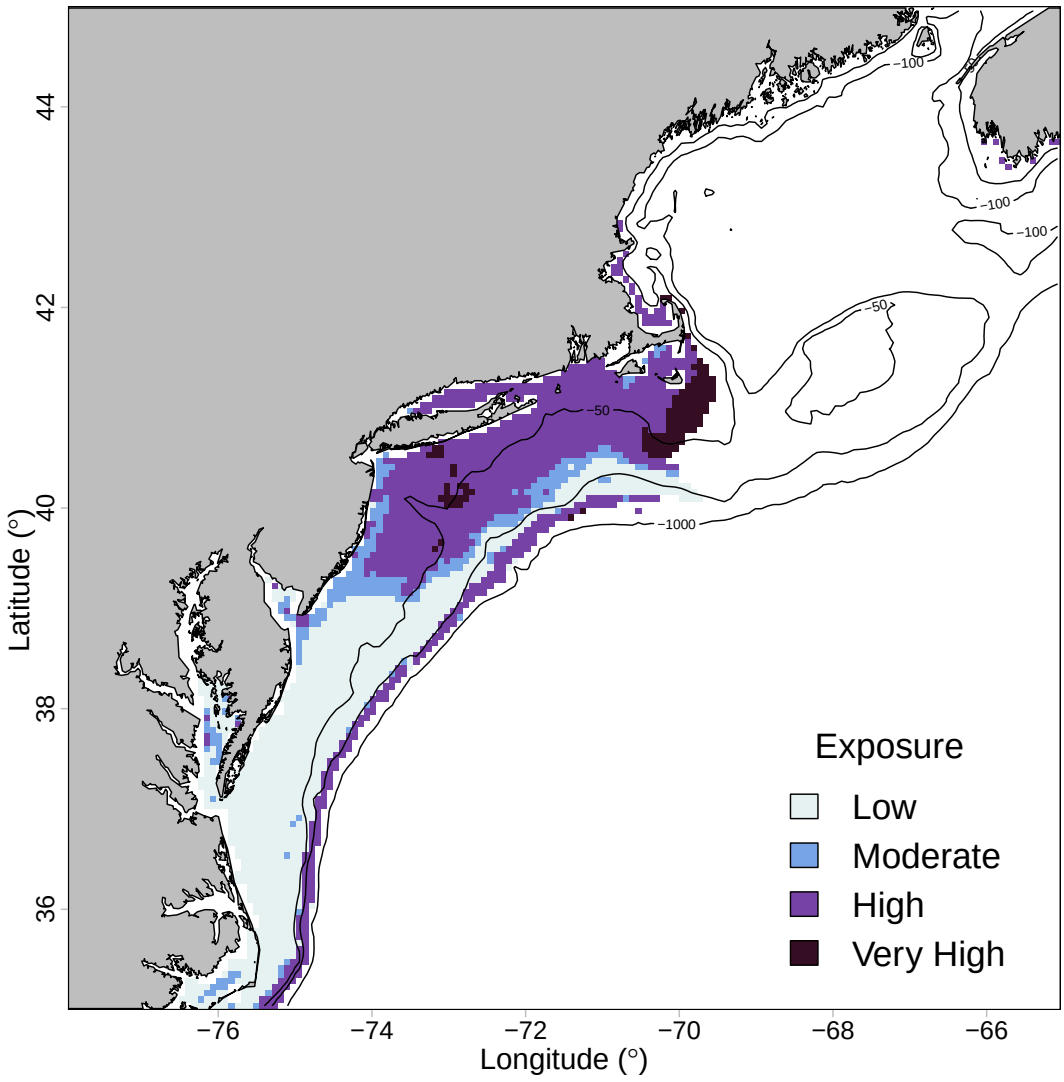


Figure 4: Total Exposure on the Northeast Shelf for Scup. Total Exposure is calculated by applying the logic rule from Hare et al. (2016) within grid cells across exposure factors (see Figure 3)

Vulnerability:

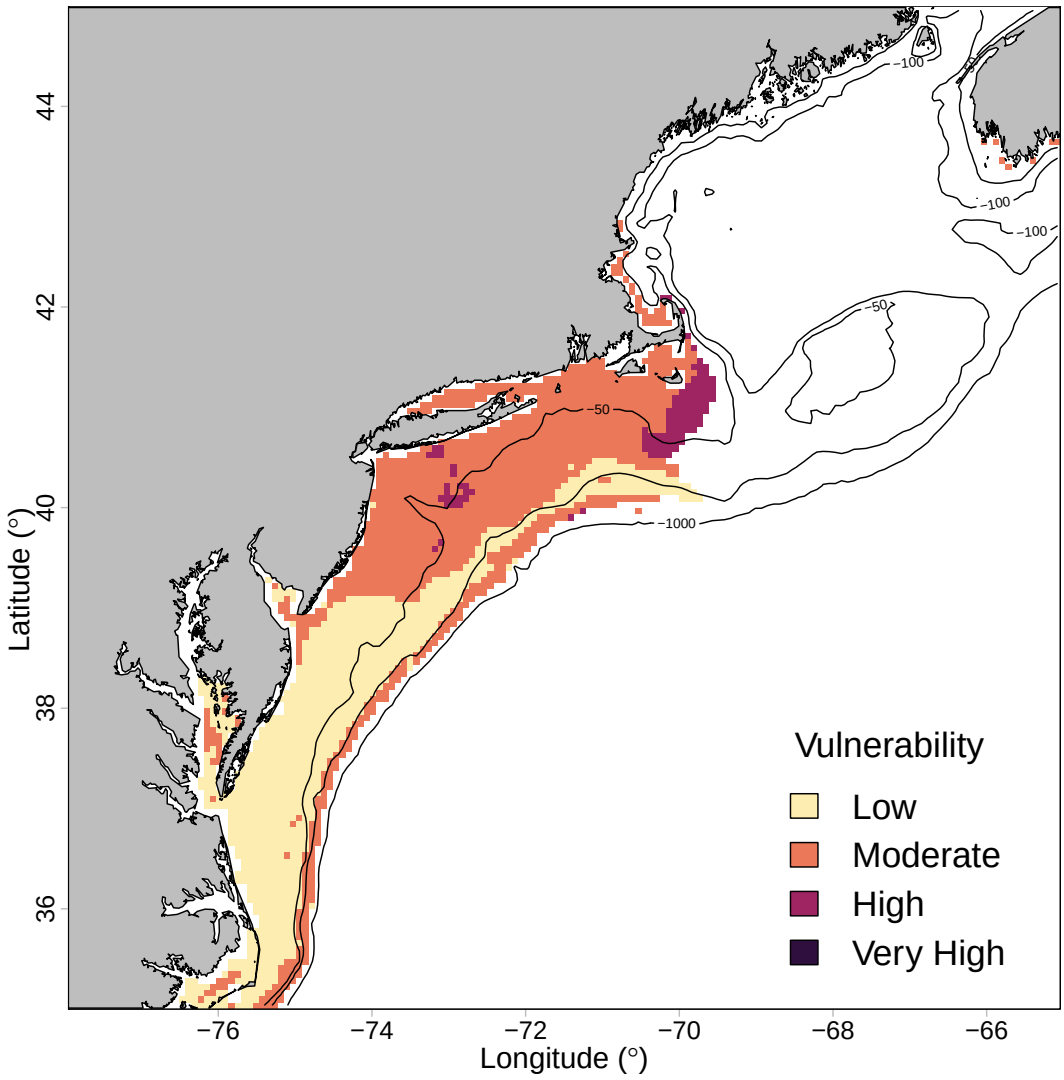


Figure 5: Vulnerability on the Northeast Shelf for Scup. Vulnerability is the product of sensitivity and exposure (see Figure 4 and Figure 2)

Overall mean Vulnerability for Scup was **Moderate** (4.15 +/- 2.03) (Figure 1). Vulnerability was Moderate in Southern New England and on the continental shelf break. It was High south of Cape Cod (Figure 5).

In the previous Assessment, Scup also received a Moderate vulnerability score (Hare et al. 2016).

Data Quality Metrics

Life History Data Quality:

98% of data quality scores were greater than or equal to 2, indicating excellent data quality. No data gaps (mean attribute data quality score < 2) were identified (Figure 1).

Model Confidence Score:

The scup ensemble model received a mean model confidence score of 2.1 (+/- 0.7), indicating moderate confidence in the model. The predicted distributions matched experts' expectations, with the exception of southern latitudes where probabilities were higher than expected in nearshore areas in the warmer months. The ensemble model AUC score was 0.95, indicating excellent model performance.

Additional Metrics

Distribution Shifts:

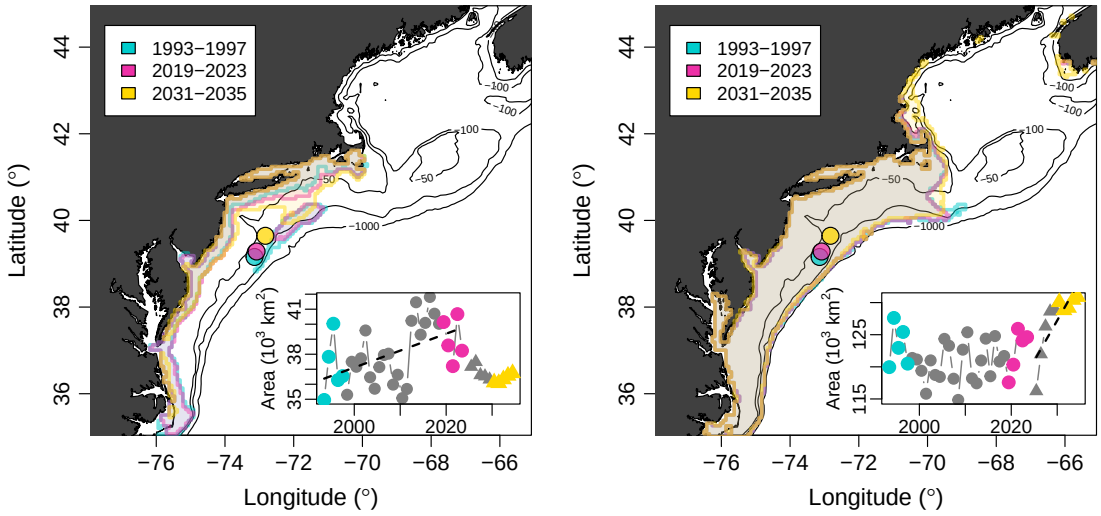


Figure 6: Distribution shifts on the Northeast Shelf for Scup. Polygons illustrate core habitat (left) and species range (right) for five-year average periods. Points illustrate weighted centroid, calculated as the average position with species distribution models as weights. Inset timeseries illustrate the shift in core and habitat range area over time. Circles represent points from the MOM6 hindcast; triangles represent points from the MOM6 forecast. Dashed lines represent significant linear regressions

Note: For an animated version of this plot, please see the online version of this document.

Despite losing core habitat on the continental shelf, scup distributions overall expanded between 1993 and 2023. Scup utilized more nearshore habitats than offshore habitats (Figure 6). The weighted centroid shifted 15 km north-northeast during this time period.

Under near future ocean conditions, scup ranges are predicted to continue their expansion. Their range is predicted to continue expanding eastward and northward into the Gulf of Maine (Figure 6). As a result, the weighted centroid is also predicted to shift 46 km northeast.

In the previous assessment, scup received a High Distributional Vulnerability rank due to their high mobility and large seasonal migrations (Hare et al. 2016).

Table 2: Distribution shifts based on five-year average distributions. Changes are calculated relative to the previous time period.

Time Period	Weighted Centroid Shift (km & bearing)	Core Habitat Area (change) (km^2)	Habitat Total Area (change) (km^2)	Latitude Range (change) (deg)	Change in Longitude Range (deg)
1993-1997	NA	37,558	123,452	5.99	5.85
2019-2023	15.0 km NNE	39,379 (+1,821)	122,333 (-1,118)	5.92 (-0.06)	5.77 (-0.08)
2031-2035	46.0 km NE	36,419 (-2,960)	129,865 (+7,531)	5.74 (-0.18)	5.85 (+0.08)

Directional Effect:

Experts predicted that climate change will have a **neutral** effect on Scup. Certainty was 93%. Scup distributions have shifted northward and offshore (Bell et al. 2015) (Palacios-Abrantes et al. 2023), likely in search of preferred temperatures, which range between 9-24C (Steimle et al. 1999), but their center of distribution has not changed (Roday et al. 2026). Juveniles and adults both utilize vulnerable estuarine habitats (Guyant et al. 2022). In the previous assessment, experts predicted that Scup would be positively impacted by climate change, however, uncertainty was high (Hare et al. 2016).

Climate Effects on Abundance and Distribution

Scup distributions in the spring and fall have shifted northward and offshore, which may necessitate new management strategies (Bell et al. 2015) (Palacios-Abrantes et al. 2023). Roday et al. (2026) analyzed fish distribution shifts using recreational catch and fishery independent data. Scup distributions showed a moderate range expansion but little to no change to center of distribution (Roday et al. 2026). Juvenile scup exposed to elevated pCO^2 in a laboratory setting had no significant effect on growth, survival, or otolith condition. Scup spawning areas in Long Island Sound are already exposed to high pCO^2, indicating some tolerance to ocean acidification (Perry et al. 2015).

Life History Synopsis

Scup are found in the Northwest Atlantic from Cape Cod, Massachusetts to Cape Hatteras, North Carolina (Steimle et al. 1999). There are two U.S. stocks: Mid-Atlantic/New England (Atlantic coast) and South Atlantic (Northeast Fisheries Science Center (NEFSC) 2025). Spawning occurs once per year in coastal bays and estuaries in May-August, beginning with lower latitudes (Steimle et al. 1999). Eggs take 2-3 days to hatch depending on temperature. The larval stage lasts from 2 mm TL to 18-19 TL; the duration of this period is unclear. Larval distributions are highest in inshore bays and estuaries, suggesting that eggs and larvae do not travel far during planktonic stages (Steimle et al. 1999). The biggest cue for larval settlement is size (~25 mm), with size in part being a function of temperature - most common at 15-20°C (Steimle et al. 1999). Larval diets are poorly understood, but larvae have been successfully reared on zooplankton in lab experiments (Bowman et al. 1987). Early juveniles consume copepods, so larval scup may also consume copepods (Michelman 1988). Larval predators include plantivorous medusae, crustaceans, and fishes (Steimle et al. 1999). Juveniles have been caught in intertidal and subtidal habitats, over sand, silty-sand, shell, mud, mussel beds and eelgrass (*Zosteria marina*). Offshore winter-spring distributions indicate that they occur in habitats ranging from relatively flat, open, sandy-silty bottoms to the head of

submarine canyons, and other areas with topographical relief and varying sediments. Juveniles can use biogenic depressions, sand wave troughs, and possibly mollusk shell fields for shelter in winter, suggesting structure may be important to juveniles (Able and Fahay 1998). Juveniles primarily feed on polychaetes, epibenthic amphipods and other small crustaceans, mollusks, and fish eggs and larvae (Steimle et al. 1999). Prey size increases with individual size, with copepods and mysids being important to early juveniles and bivalve mollusks consumed by larger fish (Bowman et al. 1987). Adult scup are also benthic feeders and forage on a variety of prey, including small crustaceans (including zooplankton), polychaetes, mollusks, small squid, vegetable detritus, insect larvae, hydroids, sand dollars, amphipods, and small fish (Steimle et al. 1999). Similar to juveniles, their prey size scales with individual size. Diets can vary based on location, and season. Feeding slows or stops entirely in winter (Byron and Link 2010). Adults are primarily benthic and utilize both estuarine and shelf benthos (Steimle et al. 1999). Age at maturity is around 2 years of age, with a maximum age between 15-20 years (Northeast Fisheries Science Center 2015). Scup undertake extensive inshore/offshore migrations driven by temperature preferences. They migrate south and offshore in the fall and return to inshore bays and estuaries in the spring to spawn. Scup can become inactive or dormant if caught in cold pockets < 6°C during their inshore spring migration and prefer temperatures 9-24°C (Steimle et al. 1999). The North Atlantic scup fishery is managed jointly by the Mid-Atlantic Fishery Management Council and Atlantic States Marine Fisheries Commission. The 2025 Management Track Assessment reported the Atlantic coast stock as not overfished and overfishing not occurring (Northeast Fisheries Science Center (NEFSC) 2025).

Supplemental Scores

Exposure Timeseries

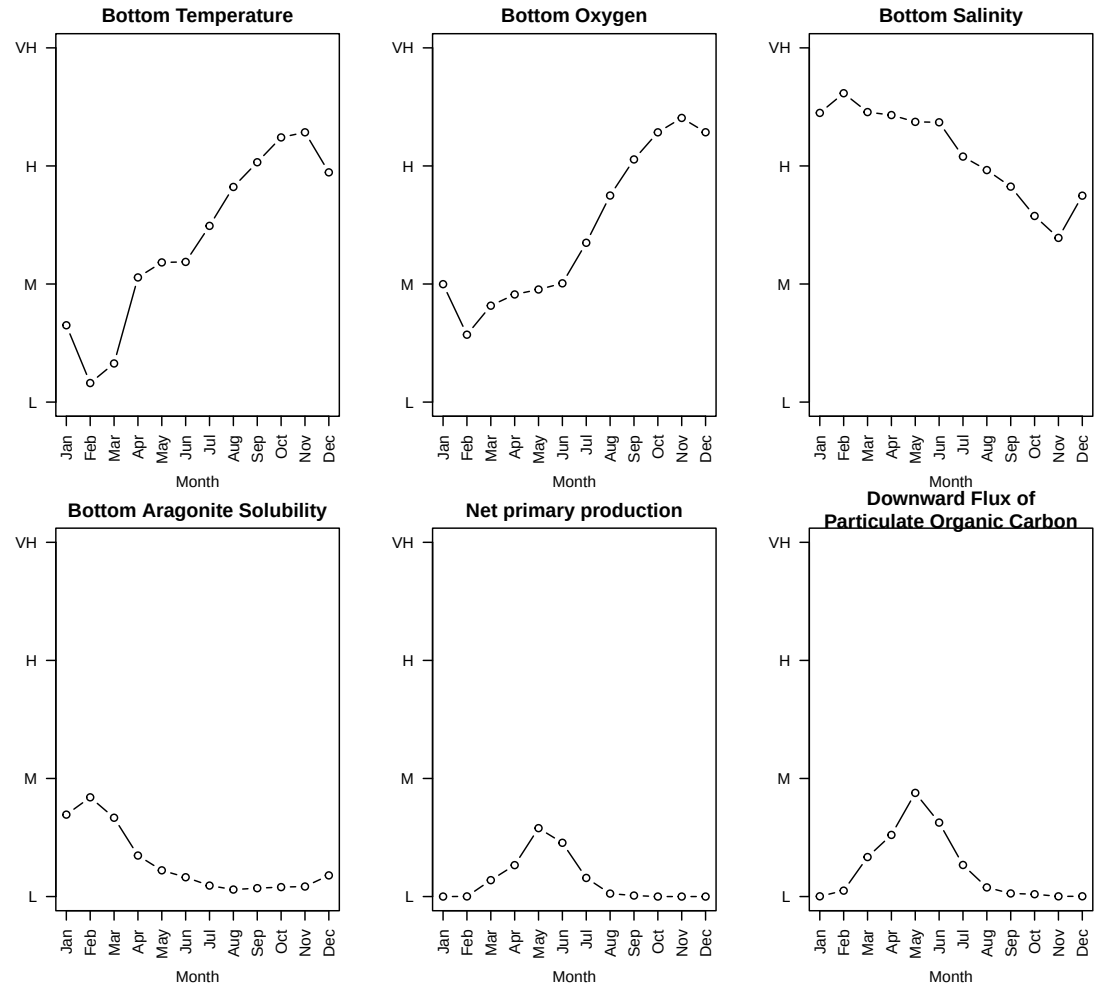


Figure 7: Monthly Timeseries of each Exposure Factor for Scup. Exposure was calculated between 2014-2023 and 2025-2035

Throughout the year, Scup exposure ranges from Low to High (Figure 8). Exposure is greatest September through November due to high bottom temperature, oxygen, and salinity exposure (Figure 7).

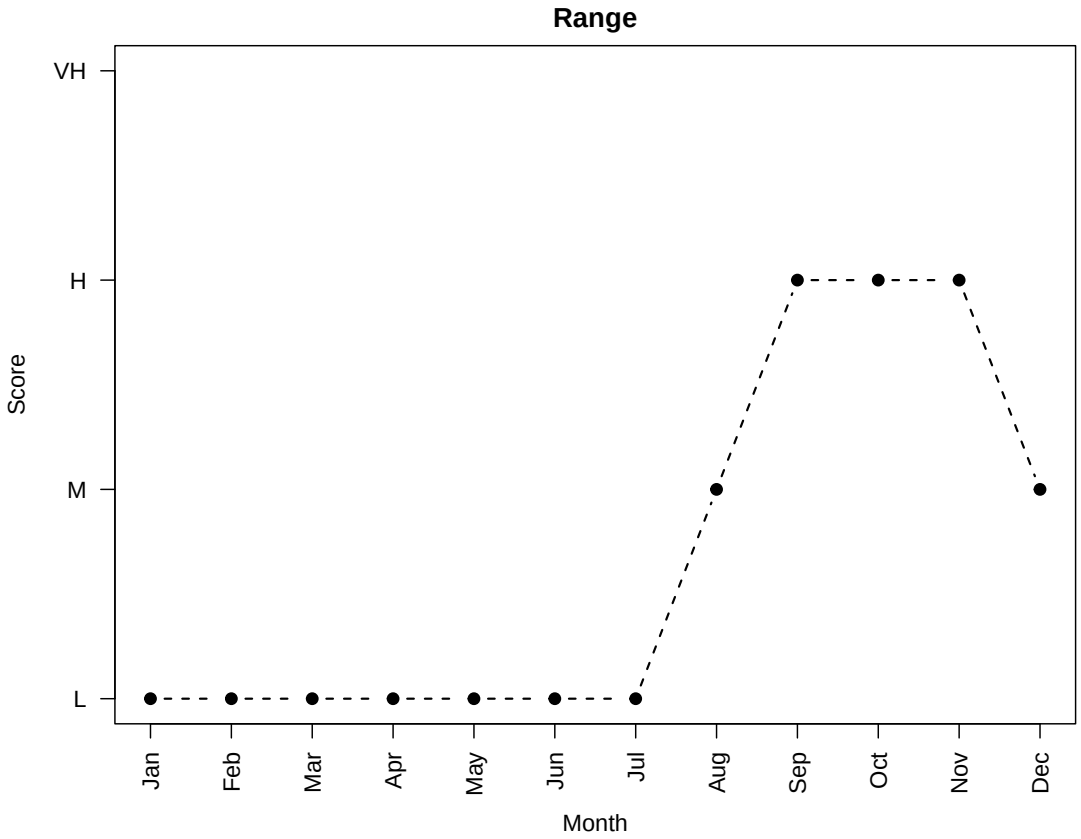


Figure 8: Timeseries of Total Exposure for Scup

Exposure due to Significant Factors

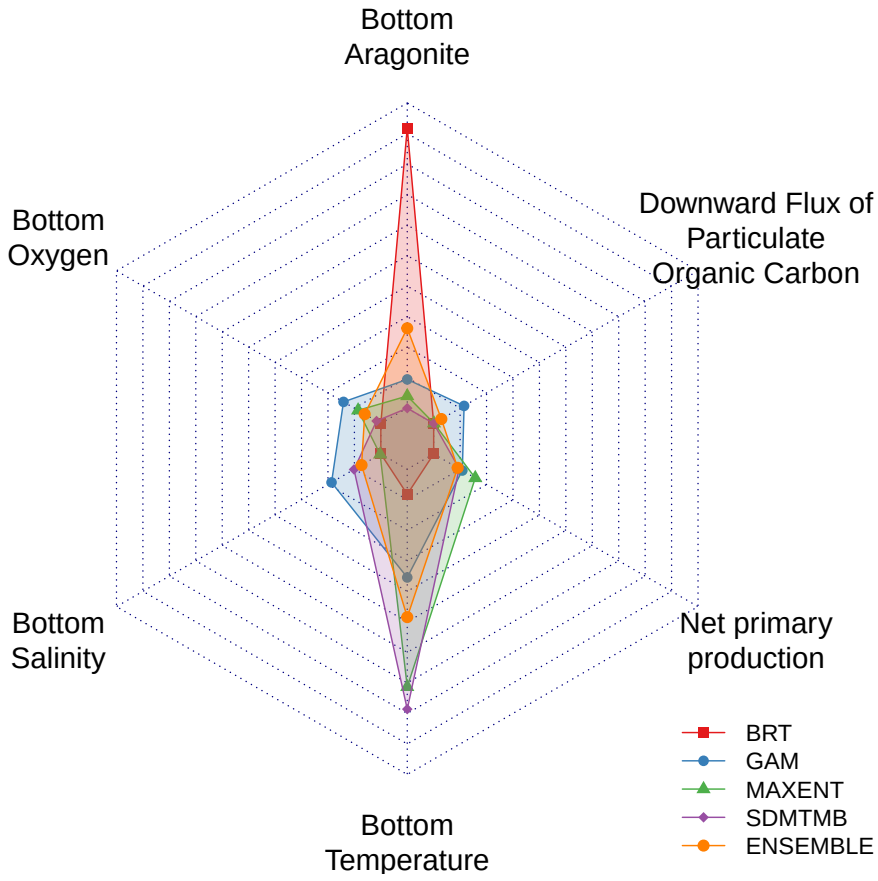


Figure 9: Dynamic Factors driving Scup distributions. The closer the polygon is to the edge of the figure, the more important the variable is. Each ring represents 10% variance explained. Factors must explain at least 10% of variance in the ensemble model to be used to calculate total exposure.

Scup distributions were driven by bottom temperature and aragonite solubility (Figure 9). Temperatures drive inshore/offshore migrations in scup (Steimle et al. 1999). Scup may also use structural habitat, including shellfish beds (Able and Fahay 1998). When only these factors were used to calculate exposure, scores were Low due to the lack of overlap in high bottom temperature and aragonite saturation state exposures (Figure 10, Figure 11).

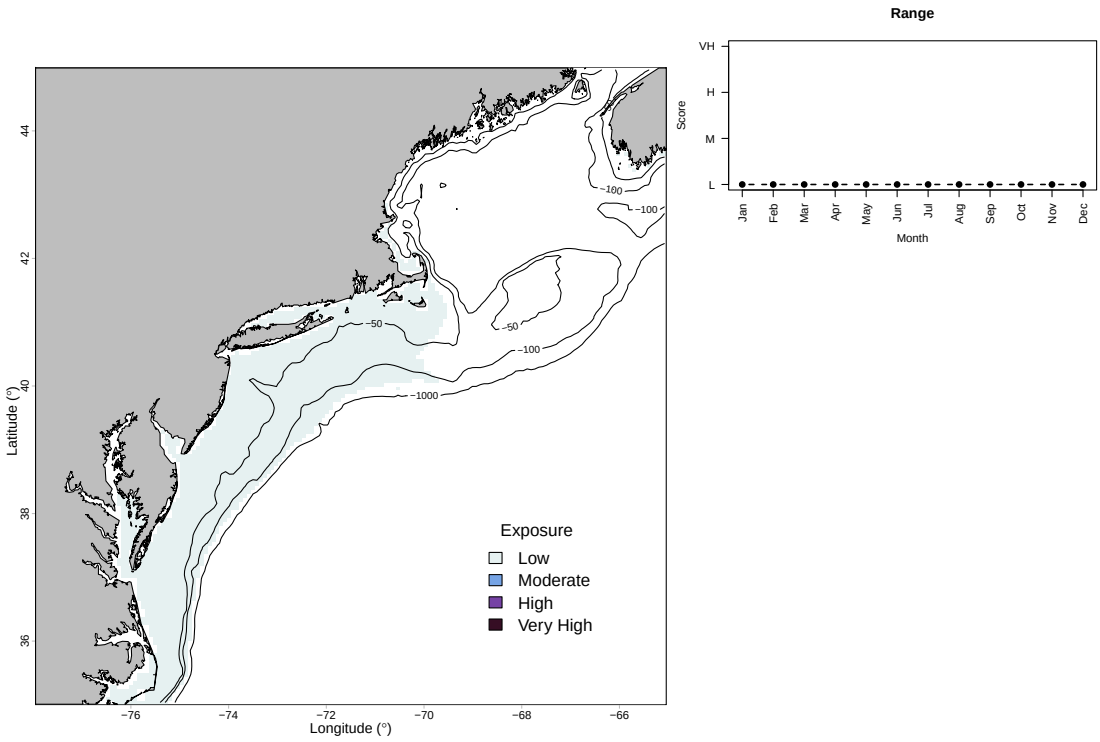


Figure 10: Total Exposure Map and Timeseries when Exposure was calculated with only Exposure Factors critical to Scup distributions

Sensitivity

Sensitivity Attributes	Score	LH DQ	Tally
Adult Mobility	1.52	2.9	
Complexity in Reproductive Strategy	2.92	2.9	
Dispersal of Early Life Stages	2.88	2.5	
Egg and Larval Survival and Settlement Requirements	2.24	2.5	
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Spawning Cycle	3	2.9	
Stock Size/Status	1.12	3	
Total Sensitivity | Certainty	**Moderate (2) | 0.59**		

Exposure

Exposure Factors	Mean	SD	Tally
Bottom Temperature	2.4	1.19	
Bottom Aragonite Solubility	1.32	0.81	
Total Exposure	**Low (1)**	**0.00**	

Vulnerability

Vulnerability	Mean	SD	Tally
Total Vulnerability	**Low (2)**	**0**	

Low Moderate High Very High

Figure 11: As in Figure 1 for Scup, but with only significant Exposure factors and the resulting Vulnerability Scores.

References

Able, Kenneth W., and Michael P. Fahay. 1998. *The First Year in the Life of Estuarine Fishes in the Middle Atlantic Bight*. Rutgers University Press.

Bell, R. J., D. E. Richardson, J. A. Hare, P. D. Lynch, and P. S. Fratantoni. 2015. "Disentangling the Effects of Climate, Abundance, and Size on the Distribution of Marine Fish: An Example Based on Four Stocks from the Northeast US Shelf." *ICES Journal of Marine Science* 72 (5): 1311–22.

Bowman, R. E., T. R. Azarovitz, E. S. Howard, and B. P. Hayden. 1987. *Food and Distribution of Juveniles of Seventeen Northwest Atlantic Fish Species, 1973-1976*. NOAA Tech. Mem. NMFS-F/NEC-45. National Oceanic; Atmospheric Administration (NOAA).

Byron, C. J., and J. S. Link. 2010. "Stability in the Feeding Ecology of Four Demersal Fish Predators in the US Northeast Shelf Large Marine Ecosystem." *Marine Ecology Progress Series* 406: 239–50.

Guyant, M, M Bachman, J Coakley, et al. 2022. *Northeast Regional Habitat Assessment-Habitat Climate Vulnerability Assessment Crosswalk, Including Species Narratives, Habitat-Species Matrix, and Final*

Report. Northeast Regional Habitat Assessment. https://nrha.shinyapps.io/dataexplorer/#!/crosswalk.

Hare, Jonathan A, Wendy E Morrison, Mark W Nelson, et al. 2016. "A Vulnerability Assessment of Fish and Invertebrates to Climate Change on the Northeast US Continental Shelf." *PloS One* 11 (2): e0146756.

Looby, Audrey, Kieran Cox, Santiago Bravo, et al. 2022. "A Quantitative Inventory of Global Soniferous Fish Diversity." *Reviews in Fish Biology and Fisheries* 32 (2): 581–95.

Michelman, M. S. 1988. "The Biology of Juvenile Scup (Stenotomus Chrysops (L.)) in Narragansett Bay, RI: Food Habits, Metabolic Rate and Growth Rate." Master's thesis, University of Rhode Island.

Northeast Fisheries Science Center. 2015. *60th Northeast Regional Stock Assessment Workshop (60th SAW) Assessment Report*. Nos. 15-08. National Marine Fisheries Service. https://doi.org/10.7289/V5W37T9T.

Northeast Fisheries Science Center (NEFSC). 2025. *Scup 2025 Management Track Assessment Report*. Atlantic States Marine Fisheries Commission. https://asmfc.org/wp-content/uploads/2025/08/Scup_Managment_Track_Assessment_2025.pdf.

Palacios-Abrantes, J., S. Crosson, C. Dumas, et al. 2023. "Quantifying Fish Range Shifts Across Poorly Defined Management Boundaries." *PLoS One* 18 (1): e0279025.

Perry, D. M., D. H. Redman, J. C. Widman Jr, S. Meseck, A. King, and J. J. Pereira. 2015. "Effect of Ocean Acidification on Growth and Otolith Condition of Juvenile Scup, Stenotomus Chrysops." *Ecology and Evolution* 5 (18): 4187–96.

Roday, R. E., R. E. Hudock, D. R. Millea, A. B. Carlisle, and E. A. Hale. 2026. "Recreational Fisheries as Indicators of Change: Species Range Shifts from MRIP (1981–2024)." *BMC Zoology*.

Steimle, F. W., C. A. Zetlin, P. L. Berrien, D. L. Johnson, and S. Chang. 1999. *Essential Fish Habitat Source Document: Scup, Stenotomus Chrysops, Life History and Habitat Characteristics*. NOAA Technical Memorandum NMFS-NE-149. National Oceanic; Atmospheric Administration (NOAA).